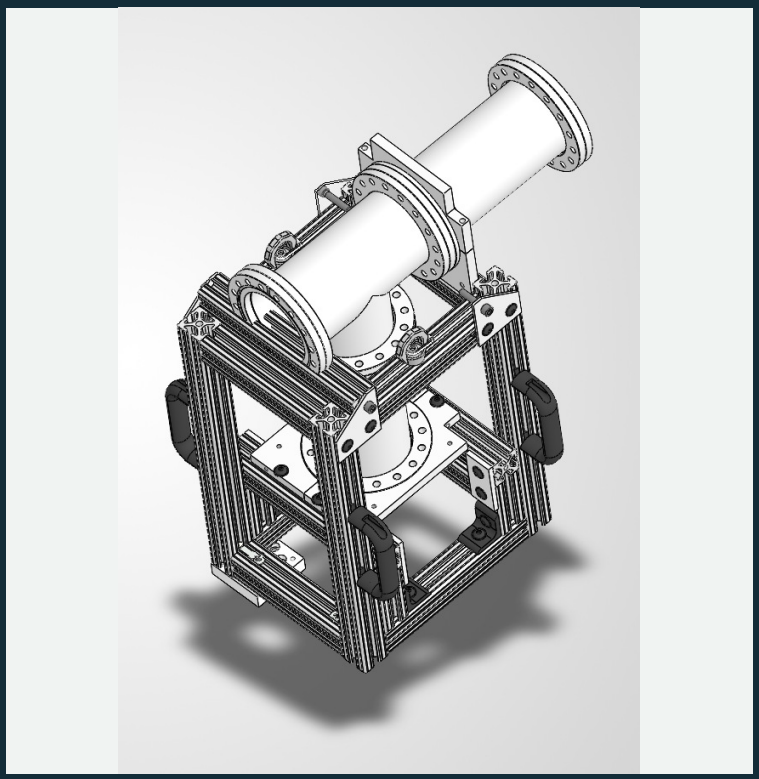
BRENDAN P. HELLMAN

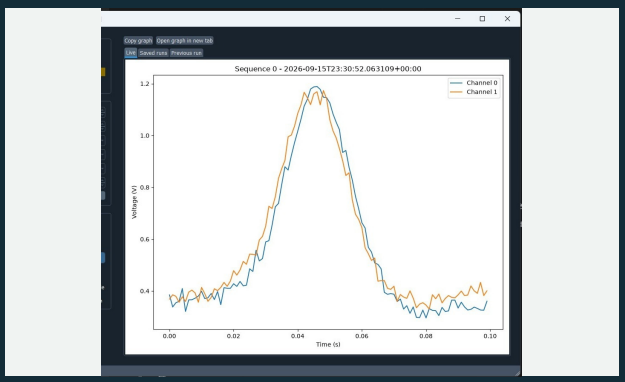
LABORATORY PROJECT PORTFOLIO

Design. Code. Build. Measure.

Laboratory hardware, scientific software, and optical systems developed through hands-on research work. From custom fixtures and instrument enclosures to acquisition interfaces, numerical models, and spectroscopy layouts.







CUSTOM HARDWARE / SCIENTIFIC SOFTWARE / OPTICAL SYSTEMS

DeMille Lab · Johns Hopkins University

Baltimore, Maryland | October 2026

bhellma22@gmail.com

Projects at a glance

Nineteen connected project stories, organized by the work involved. Build notes, code, measured records, design files, and planning documents are identified within each case study.

01	**Source-region mounting systems**	Mechanical
02	**Alpha-detector stand**	Mechanical
03	**60 Hz trigger box**	Electronics
04	**Optical shutter mounts**	Mechanical
05	**Shutter timing characterization**	Analysis
06	**Coaxial-relay rack enclosure**	Mechanical
07	**Dual-buffer enclosure integration**	Mechanical
08	**PVD chamber print mockup**	Mechanical
09	**Analog-input acquisition GUI**	Software
10	**Universal scientific-camera GUI**	Software
11	**Lab-control integration and documentation**	Software
12	**Extensible experiment-control prototype**	Software
13	**NI-DAQmx hardware test utility**	Software
14	**MOT-loading models and fitting**	Analysis
15	**Cryostat absorption-spectroscopy layout**	Optics
16	**Laser collimation and characterization**	Optics
17	**Laser feedforward circuit**	Electronics
18	**Laser-diode protection PCB**	Electronics
19	**Crystal-spectroscopy platform planning**	Research

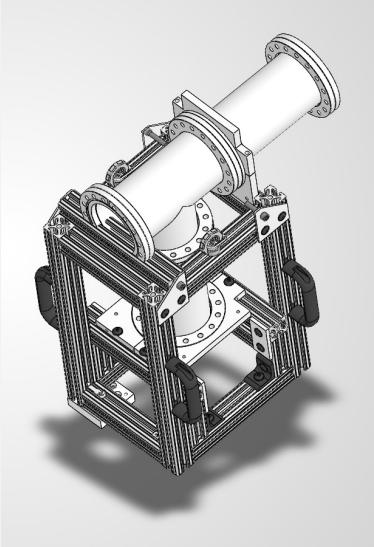
Each project has a page in this PDF. The offline edition adds search, image galleries, and the full source register. Click any project above to jump to its page.

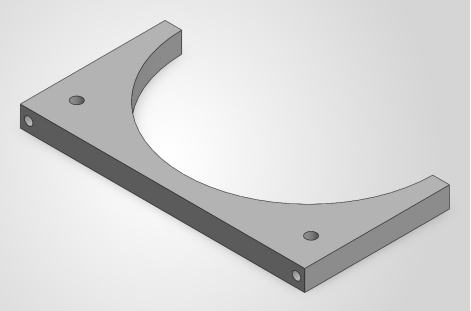
MECHANICAL / PROJECT 01

Source-region mounting systems

BUILT PROTOTYPE + CAD

A modular support structure for the lab's source-region oven, with a smaller prototype assembly and a larger frame design. The work connects equipment layout, custom flange supports, standard extrusion hardware, and a practical assembly sequence.





Saved SolidWorks model of the mounted prototype oven. Additional view: Custom slim lower flange support; CAD view.

CONTRIBUTION

- Modeled the mounting assemblies and revised the support geometry to meet space constraints.
- Developed slim CF600 flange clamps and modified 80/20 plates, with waterjet drawings and drilling/tapping details.

DELIVERABLES

Native SolidWorks assemblies and STEP exports · Custom-part drawings, parts lists, and assembly instructions

OUTCOME

The prototype mounting notes describe the oven installed on the frame and record a useful build revision: the second set of lower flange clamps was unnecessary in practice.

TOOLS & METHODS

SolidWorks · 80/20 framing · Waterjet · DFM · BOM

1/4 in
Plate stock in build notes

1/2 in
Flange-clamp stock

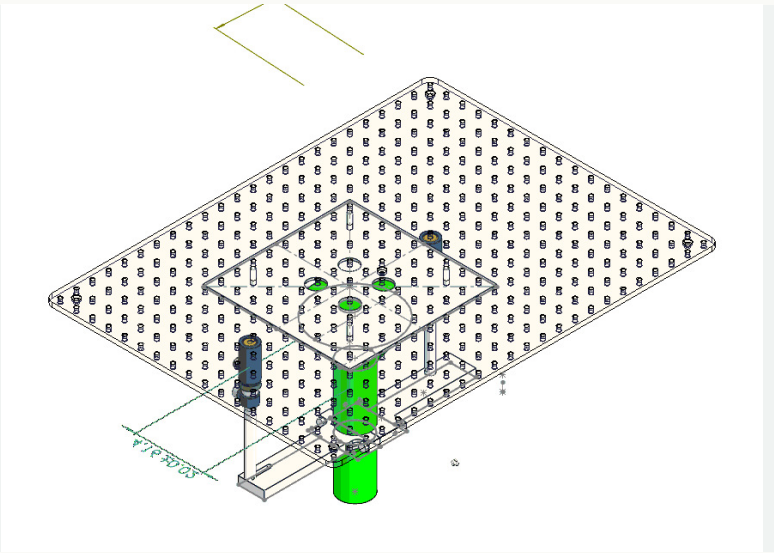
The installed prototype is documented in the assembly notes. The larger source-region frame is presented as a CAD design; no load-rating or structural certification is claimed.

MECHANICAL / PROJECT 02

Alpha-detector stand

BUILT / DOCUMENTED

A breadboard-mounted fixture that holds an alpha detector and positions a filter-paper sample. Custom split clamps connect the detector to standard optical posts while a separate sheet-metal stand locates the sample.



Saved detector-stand assembly view; commercial breadboard and posts are integrated with custom supports.

CONTRIBUTION

- Designed the detector clamps, sample stand, and complete assembly.
- Prepared waterjet drawings and adjusted the clamp fit after the first version.

DELIVERABLES

SolidWorks parts and assembly · DXF drawing revisions and assembly/BOM notes

OUTCOME

The build notes record a fit correction of 0.01 in on the clamp mating faces and distinguish the posts modeled in CAD from the 6 in posts used in the physical setup.

TOOLS & METHODS

SolidWorks · Waterjet · Drill press · Tapping · Fixture design

1.89 in
Detector shaft in notes

0.01 in
Documented fit revision

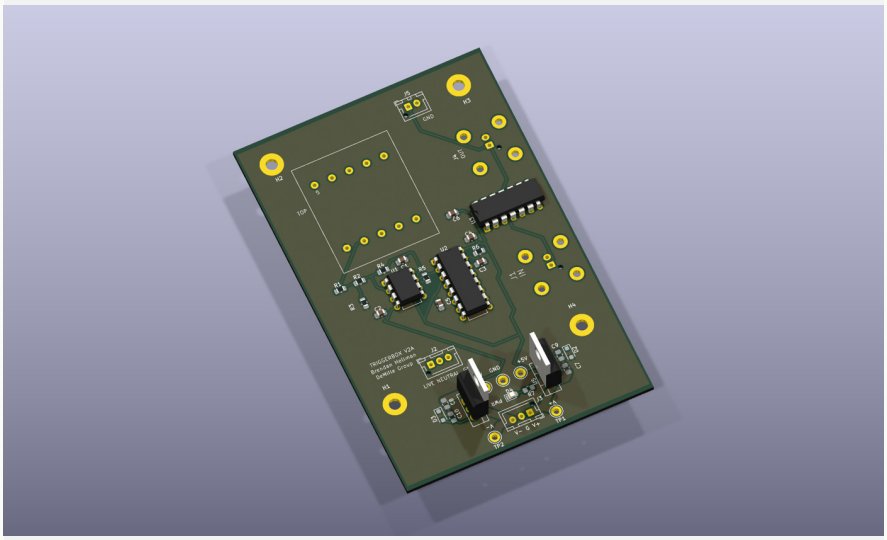
The source notes describe the completed setup. Its referenced build photograph is not included in this source-folder snapshot.

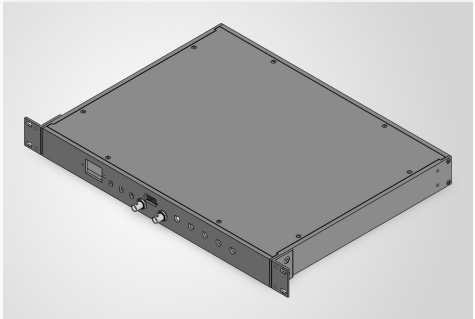
ELECTRONICS / PROJECT 03

60 Hz trigger box

BUILT / DOCUMENTED

A laboratory interface that turns a 60 Hz reference and an enable input into active-low trigger pulses. The project spans the circuit, PCB revisions, rack enclosure, connector layout, wiring, and fabrication documentation.





3D rendering of the saved trigger-board layout; several connector and transformer models are absent. Additional view: Saved rack-enclosure CAD view.

CONTRIBUTION

- Developed the trigger-circuit and enclosure revisions, including panel cutouts and board-to-panel connector spacing.
- Documented the transformer, comparator, one-shot, and NAND stages, plus the box wiring and assembly sequence.

DELIVERABLES

KiCad schematic, PCB, and production BOM · SolidWorks enclosure assemblies and waterjet DXFs · Circuit and box assembly guide

OUTCOME

The documented design produces a nominal 100 microsecond low pulse per selected 60 Hz edge while enabled. The guide captures corrections to connector cutouts and an earlier silkscreen ground label.

TOOLS & METHODS

KiCad · SolidWorks · PCB layout · Soldering · BNC / D-sub wiring

60 Hz
Reference repetition rate

100 µs
Nominal pulse width

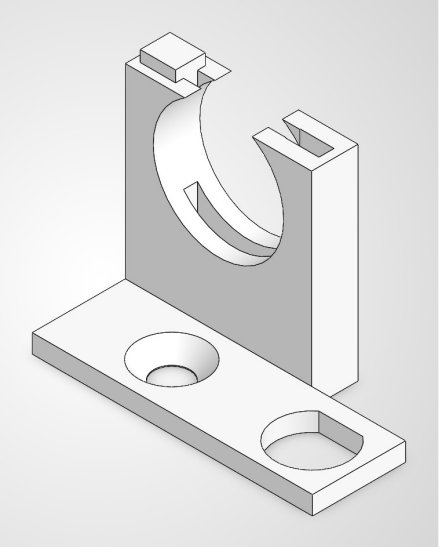
The design uses standard comparator, monostable, NAND-gate, and regulator building blocks. Pulse width is a documented design target, not a measurement newly verified for this portfolio.

MECHANICAL / PROJECT 04

Optical shutter mounts

FABRICATION DOCUMENTED

A compact, printable mount that holds a camera shutter on an optical post and provides space for a BNC connector. Multiple saved versions show the progression toward the final mount geometry.



Saved v10 shutter-holder CAD view. The shutter and BNC connector are separate components.

CONTRIBUTION

- Iterated shutter-holder geometry through the saved v6-v10 revisions.
- Integrated an 8-32 post interface, shutter slot, and side-connector cutout.

DELIVERABLES

SolidWorks v10 part · 3MF print file, earlier STL variants, and installation guide

OUTCOME

The final guide specifies basic PLA on a Bambu X1 Carbon and records the shutter's installation orientation. The design packages the shutter as a reusable optical-table component.

TOOLS & METHODS

SolidWorks · 3D printing · PLA · Optomechanics

8-32
Optical-post screw

v10
Final documented revision

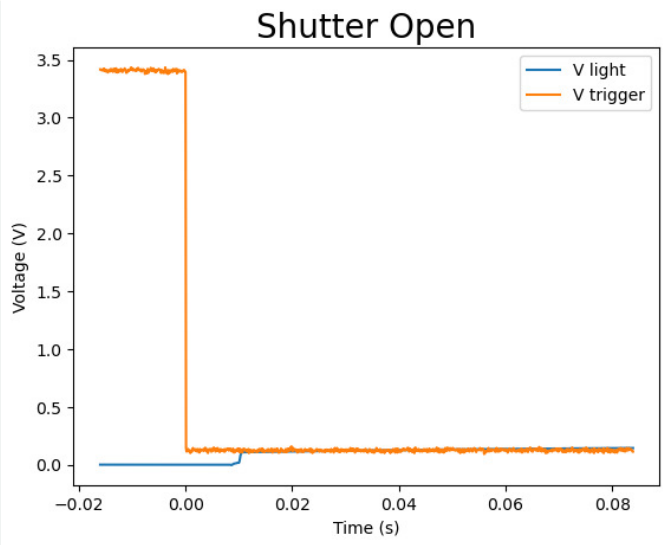
The Waterloo shutter design is retained in the archive as a reference. This case study focuses on the saved custom mount revisions. The final guide references installed-shutter photographs that are absent from this folder snapshot.

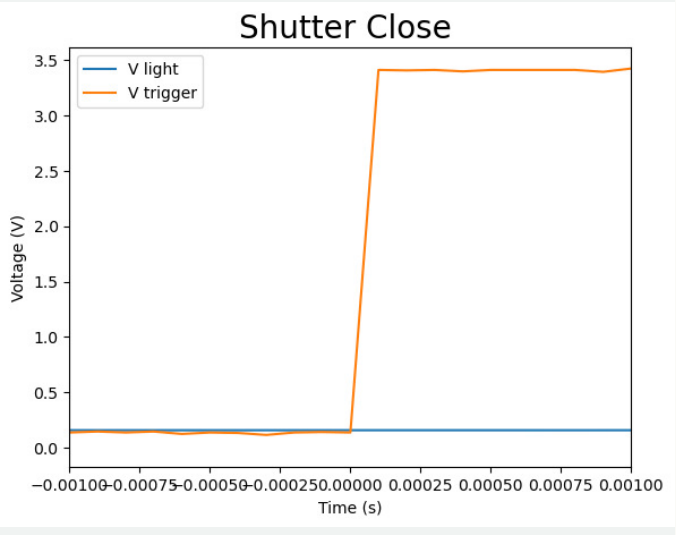
ANALYSIS / PROJECT 05

Shutter timing characterization

MEASURED / NOTEBOOK

An oscilloscope-to-Python workflow for comparing the shutter command with an optical signal. The notebook reads opening and closing traces, plots the two channels, and records the shutter's response timing.





Opening-response plot extracted from the saved notebook; full measurement window. Additional view: Closing-response plot extracted from the saved notebook; the saved view is tightly zoomed.

CONTRIBUTION

- Processed Rigol CSV traces in a Jupyter notebook using pandas, NumPy, and Matplotlib.
- Recorded opening and closing delays using the notebook's stated optical-signal thresholds.

DELIVERABLES

Jupyter analysis notebook · RigolDS0.csv and RigolDS20.csv scope exports · Saved opening and closing plots

OUTCOME

The notebook reports a 9.41 ms opening delay and 1.06 ms rise time; closing is reported as an 8.05 ms delay and 1.40 ms fall time. These are notebook-reported results, with no uncertainty or repeatability study supplied.

TOOLS & METHODS

Python · Jupyter · pandas · Oscilloscope · Timing analysis

9.41 ms
Reported opening delay

8.05 ms
Reported closing delay

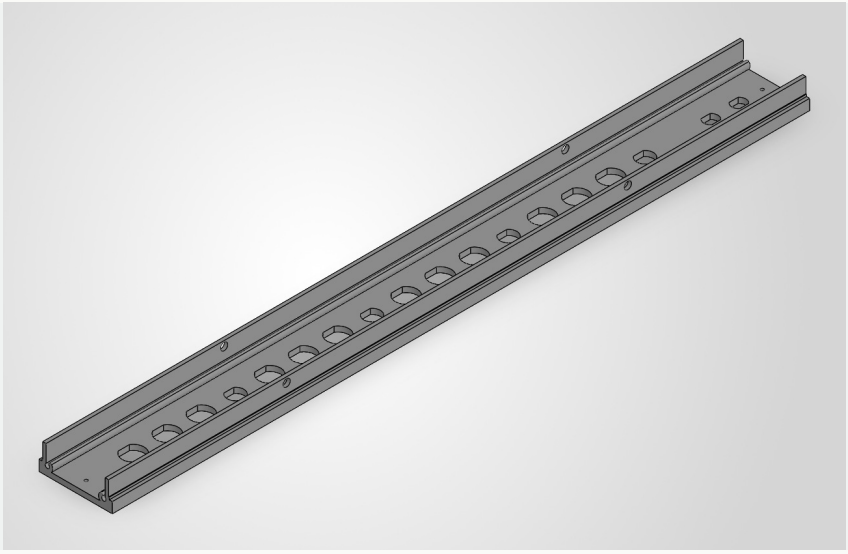
Timing values are transcribed from the notebook narrative. They were not recalculated or treated as calibrated specifications.

MECHANICAL / PROJECT 06

Coaxial-relay rack enclosure

CAD + FABRICATION FILES

A custom front-panel and mounting design for an electromechanical coaxial-relay box. The file set carries the enclosure from a SolidWorks part to shop drawings and waterjet-ready geometry.



Saved CAD view of the relay-box front and mounting geometry.

CONTRIBUTION

- Modeled a 1U front-panel variant with connector openings and mounting features.
- Prepared alternate drawing versions, separate mounting-piece geometry, and cutting files.

DELIVERABLES

Native SolidWorks part and drawings · DXF and OMAX machining files · Enclosure BOM workbook

OUTCOME

The archive preserves fabrication-oriented variants with and without mounting holes, making the mechanical choices and revision history reviewable.

TOOLS & METHODS

SolidWorks · Rack hardware · DXF · Waterjet preparation

1U

Rack-format design

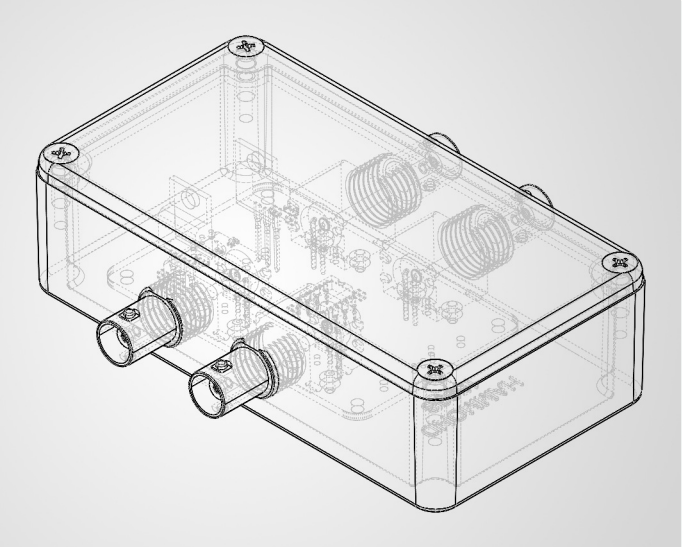
Fabrication files are present; completed assembly and electrical testing are not established by this folder.

MECHANICAL / PROJECT 07

Dual-buffer enclosure integration

CAD INTEGRATION

Mechanical packaging for a laboratory dual bipolar buffer board. The enclosure assembly brings the existing PCB and connectors into a compact Hammond-style instrument box.



Saved enclosure assembly view. Transparency reveals the packaged board and connector positions.

CONTRIBUTION

- Modeled the board and enclosure together to review the packaging and connector interfaces.
- Retained the board's source documentation alongside the mechanical integration files.

DELIVERABLES

Boxed Bipolar Buffer.SLDASM · Board assembly and STEP model · Existing circuit documentation and revision notes

OUTCOME

The saved assembly provides a mechanical integration reference for the buffer electronics. The underlying unity-gain circuit and its historical scope results are credited to the original lab design.

TOOLS & METHODS

SolidWorks · PCB packaging · Enclosure integration

2 channels

Inherited buffer-board design

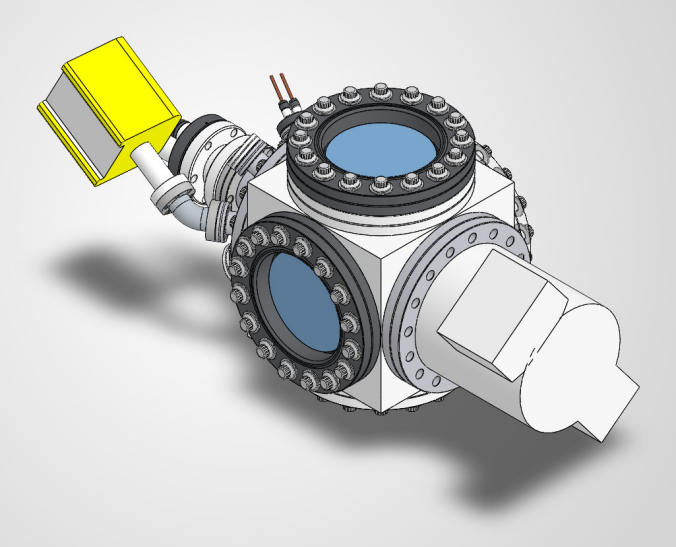
Original buffer PCB, circuit documentation, and historical characterization: Wesley Cassidy (repository history, 2023-2024). This portfolio claims enclosure integration, not authorship of the original circuit or its historical measurements.

MECHANICAL / PROJECT 08

PVD chamber print mockup

CAD + PRINT PREPARATION

A physical-scale mockup workflow for a PVD chamber assembly. The source set contains the complete CAD, a part representation, and separate left- and right-side STEP exports for downstream fabrication or printing.



Saved PVD mockup CAD view; a model of the chamber geometry, not a photograph of a print.

CONTRIBUTION

- Prepared a chamber mockup assembly and a derived part file.
- Separated the geometry into left and right exports, including a right-side variant without the end.

DELIVERABLES

3D printed PVD mockup.SLDASM · Native part and three sectional STEP exports

OUTCOME

The available deliverable is a set of CAD and split geometry files suitable for reviewing the chamber layout and preparing a physical mockup.

TOOLS & METHODS

SolidWorks · STEP · Model preparation · 3D printing

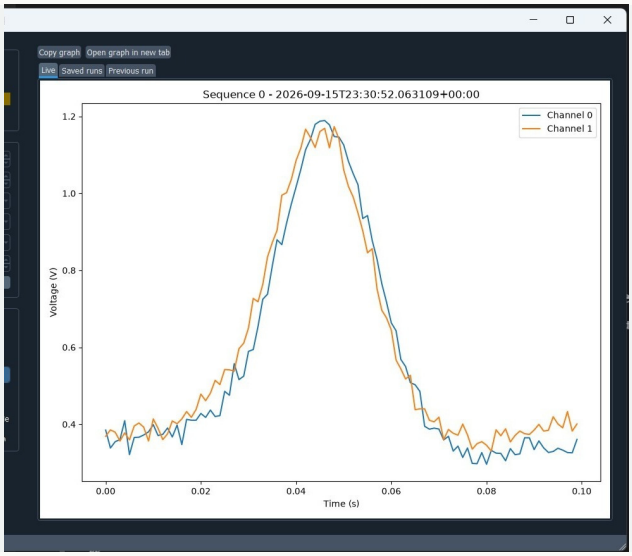
The folder does not establish completion of the physical print or operation of the represented chamber.

SOFTWARE / PROJECT 09

Analog-input acquisition GUI

IMPLEMENTED / DOCUMENTED

A PyQt5 application for externally triggered MCC analog acquisition, with a standalone manual workflow and TCP integration with the experiment-control software. The interface follows a trace from arming through storage, live display, and later review.



Saved application capture with the dummy MCC backend and synthetic two-channel traces.

CONTRIBUTION

- Separated acquisition, device backends, storage, history, plotting, and configuration into dedicated modules.
- Added threaded acquisition, HDF5/CSV output, saved-run browsing, unit controls, and explicit real-versus-dummy backend selection.

DELIVERABLES

Python application and automated-test source · README, GUI captures, and configuration templates

OUTCOME

The documented workflow arms the real DAQ and waits for a physical trigger. A restricted dummy backend supports development on unrecognized computers; the portfolio capture shows synthetic traces.

TOOLS & METHODS

Python · PyQt5 · MCC USB-231 · TCP · HDF5 · CSV

HDF5 + CSV
Trace storage

TCP
Sequence metadata

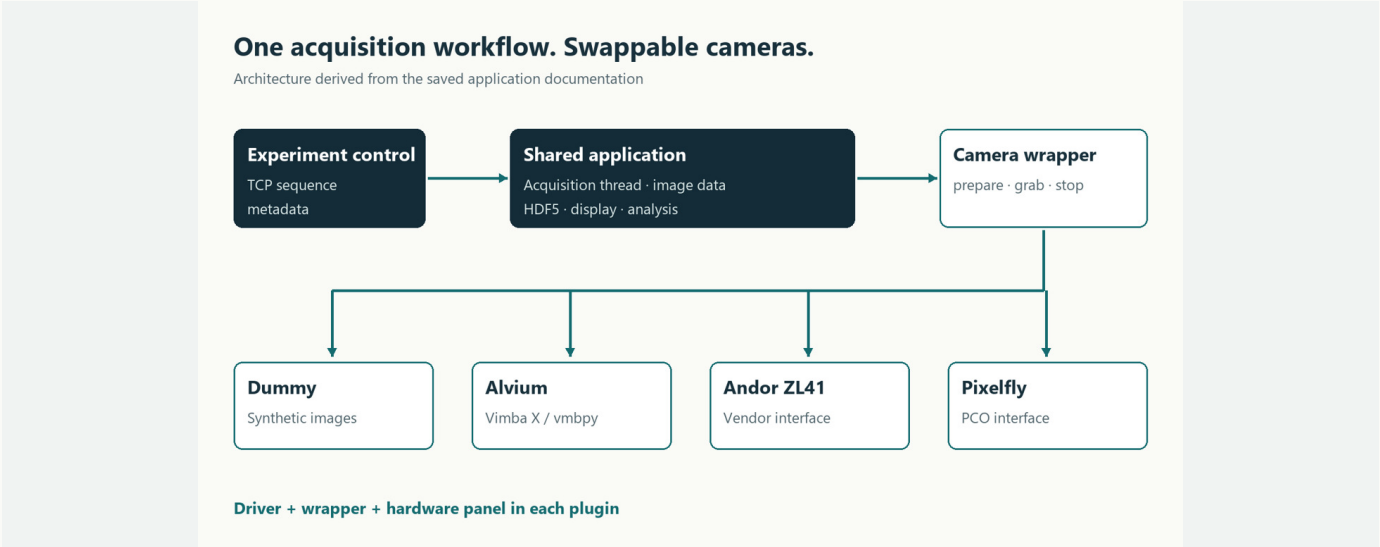
Application features are supported by code and documentation. No new hardware-acquisition or test-suite result is claimed here.

SOFTWARE / PROJECT 10

Universal scientific-camera GUI

IMPLEMENTED / DOCUMENTED

A camera-agnostic imaging application that shares experiment sequencing, display, saving, and analysis across multiple camera backends. Camera-specific SDK behavior is isolated behind a common plugin interface.



Portfolio diagram derived from the saved camera-application README; this is an architecture diagram, not a GUI capture.

CONTRIBUTION

- Organized each camera plugin into a driver, acquisition wrapper, and settings panel.
- Kept TCP sequencing, HDF5 saving, Gaussian fitting, atom-number estimation, and scan plotting in the shared workflow.

DELIVERABLES

PyQt5 application and camera plugin packages · Dummy, Alvium, Andor ZL41 Wave, and Pixelfly interfaces · Architecture and plugin-extension documentation

OUTCOME

The documented wrapper contract uses prepare_sequence(), grab_frame(), and stop_sequence(). New camera folders with hardware configuration files are discovered by the launcher.

TOOLS & METHODS

Python · PyQt5 · Scientific imaging · Plugin architecture · HDF5

3 methods
Shared acquisition contract

4 backends
Including the dummy camera

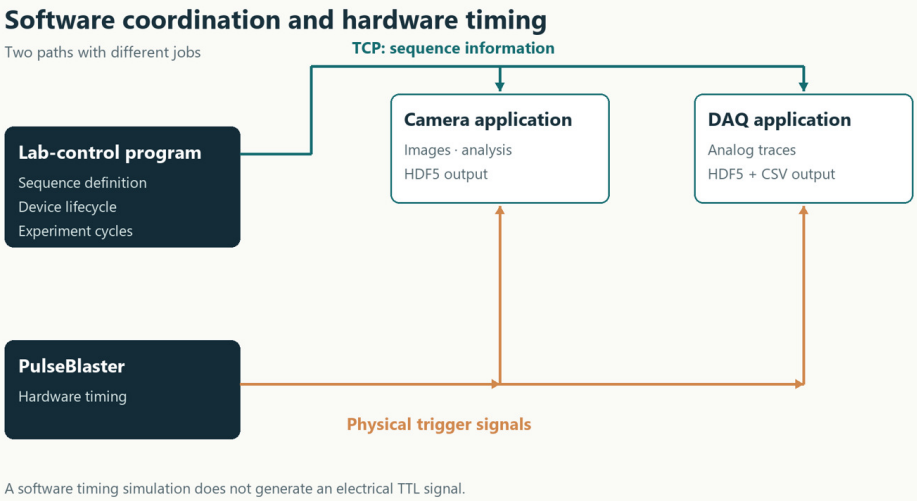
Scientific-camera SDKs and vendor drivers remain vendor components; the case study focuses on the shared application and integration architecture. The presence of a camera plugin does not establish that every hardware model was tested.

SOFTWARE / PROJECT 11

Lab-control integration and documentation

INTEGRATION / DOCUMENTATION

Work within an existing laboratory-control codebase to connect sequence orchestration, camera acquisition, analog acquisition, and hardware timing. The archive includes the inherited controller, integration modules, and substantial system documentation.



Portfolio diagram of the documented software-message and physical-trigger paths.

CONTRIBUTION

- Documented the existing module inventory and the relationship between experiment cycles, device lifecycle methods, and acquisition programs.
- Worked on the interfaces between the main controller and the DAQ/camera applications.

DELIVERABLES

Fr-lab-control source variants · Laboratory-control software reference and module inventory PDFs · DAQ and camera sequence/TCP documentation

OUTCOME

The documented architecture distinguishes software messages that carry sequence information from the physical timing signals used by laboratory hardware.

TOOLS & METHODS

Python · PyQt5 · PulseBlaster · TCP integration · Technical documentation

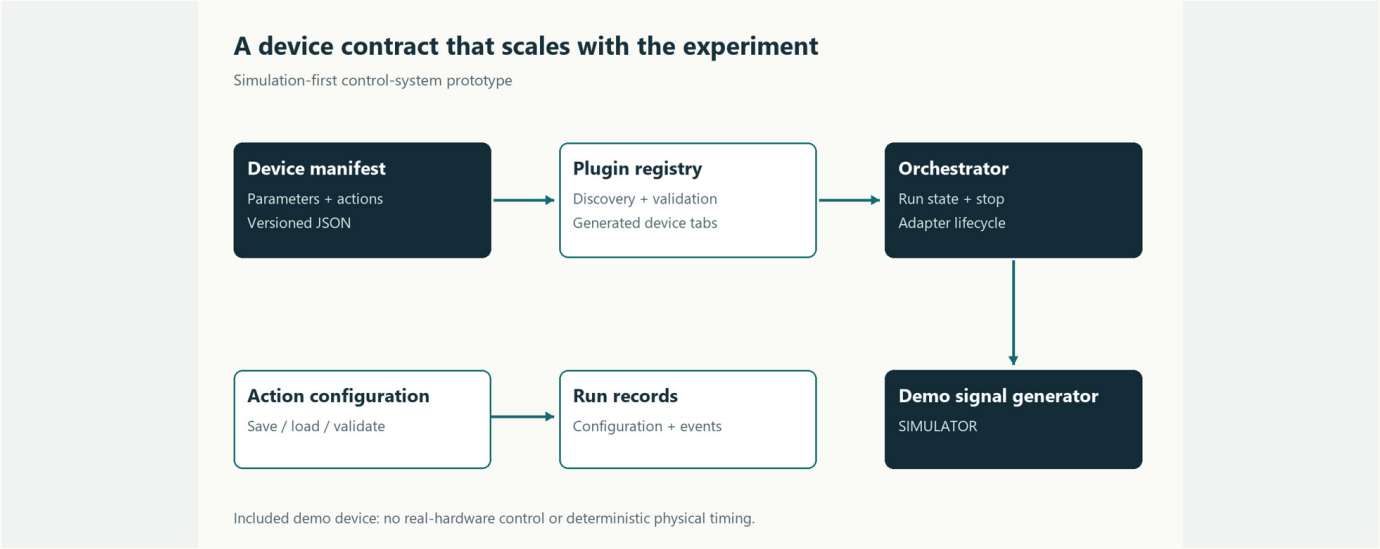
Fr-lab-control / SrF-Lab-Control is an inherited lab codebase with prior contributors. This case study describes integration and documentation work. Software simulation is not electrical TTL output or a hardware-timing validation.

SOFTWARE / PROJECT 12

Extensible experiment-control prototype

SIMULATION PROTOTYPE

A newer control-system foundation that discovers devices from manifests and runs validated action configurations through a GUI-independent orchestrator. The included signal generator is a simulator used to develop the extension pattern.



Portfolio diagram of the implemented manifest-to-orchestrator design.

CONTRIBUTION

- Defined device manifests and a common adapter lifecycle so the GUI can create device tabs from metadata.
- Implemented configuration validation, save/load, run records, stop handling, and explicit orchestration states.

DELIVERABLES

Python desktop prototype and Windows launcher · JSON schemas, example sequence, tests, and plugin-authoring guide · 23-page main guide in the source archive

OUTCOME

The source provides a complete simulation-first path for adding a device plugin, validating an action configuration, and recording a run without physical hardware.

TOOLS & METHODS

Python · JSON schema · State machines · Plugin discovery · Simulation

JSON
Device and action schemas

Simulator
Included demonstration device

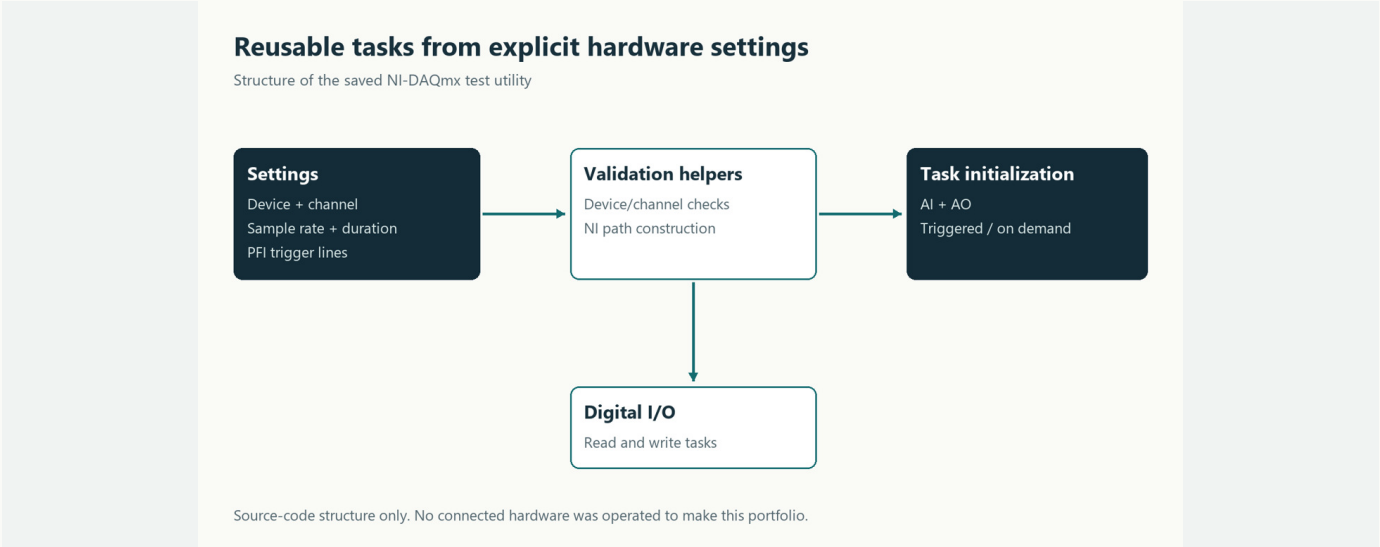
The README explicitly identifies a simulation-first skeleton. Real-hardware control and deterministic timing are not implemented by the included demonstration device.

SOFTWARE / PROJECT 13

NI-DAQmx hardware test utility

CODE / BENCH-TEST UTILITY

A parameterized Python utility for exercising National Instruments analog and digital channels. Device names, channel names, PFI trigger lines, sample rate, and acquisition duration are grouped as settings.



Portfolio diagram of the test utility's channel and task structure.

CONTRIBUTION

- Implemented reusable initializers for triggered and on-demand analog input/output tasks.
- Added channel-path helpers and validation for the supported device/channel selections, plus digital I/O routines.

DELIVERABLES

NIDAQ code test.py · NI DAQ test.xlsx measurement workbook

OUTCOME

The code separates hardware configuration from task construction, making the same test structure reusable across Dev1/Dev2 and the selected input/output channels.

TOOLS & METHODS

Python · NI-DAQmx · Analog I/O · Digital I/O · Triggering

AI / AO
Triggered or on demand

PFI
Configurable trigger paths

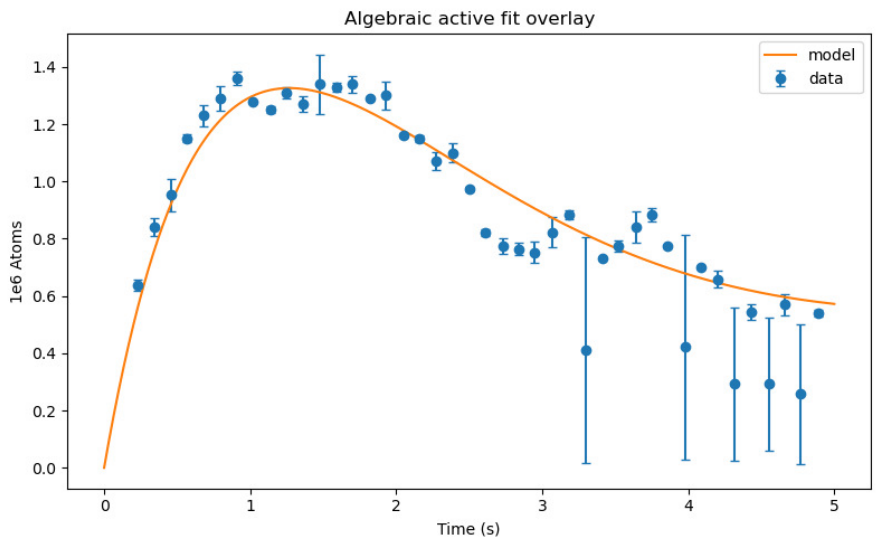
This case study describes the saved test code; no new task was executed against connected laboratory hardware.

ANALYSIS / PROJECT 14

MOT-loading models and fitting

CODE + SAVED ANALYSIS

Python translations of a Mathematica MOT-loading model, combining differential-equation solving with weighted nonlinear fitting. Saved notebooks preserve model overlays and residual plots for both algebraic and coupled-ODE approaches.



Cached fit overlay from the saved MOT-loading notebook; the original input dataset was not reloaded for this portfolio.

CONTRIBUTION

- Translated data scaling, model definitions, and numerical integration into Python/SciPy.
- Implemented weighted least-squares fits, covariance-based uncertainties, confidence intervals, and residual analysis, including an eight-parameter fit variant.

OUTCOME

The work makes the modeling workflow inspectable in Python and preserves the distinction between model assumptions, fitted curves, and residual structure.

TOOLS & METHODS

Python · SciPy · NumPy · Jupyter · ODEs · Weighted fitting

DELIVERABLES

Python scripts and Jupyter notebooks · Cached model/fit and residual figures

8 parameters
Dedicated full-fit script

ODE + fit
Numerical model workflow

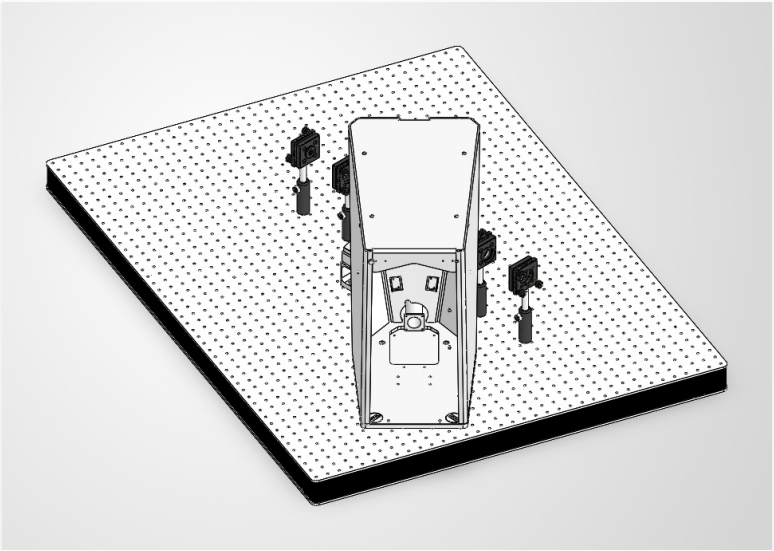
The scripts explicitly translate an existing Mathematica notebook. Model provenance is retained; the portfolio does not claim original authorship of the underlying physics model. Figures are saved notebook outputs, not a newly reproduced analysis. Experimental validity and parameter identifiability are not established by the plots alone.

OPTICS / PROJECT 15

Cryostat absorption-spectroscopy layout

CAD LAYOUT

An optical-breadboard layout for near-1115 nm absorption spectroscopy through a cryostat. The assembly integrates fiber-launch and receive packages, lenses, adjustable mounts, posts, and the cryostat's mechanical envelope.



Saved spectroscopy assembly view; the cryostat and commercial optical hardware are layout references.

CONTRIBUTION

- Arranged the launch, focusing, sample, collection, and receive stations around a common beam axis.
- Revised lens and support choices to clear the cryostat and documented the retained/suppressed design alternatives.

DELIVERABLES

SolidWorks assembly revisions and beam-path SVG · V1b optical layout notes and component list · Saved placement, interference, and reopen logs

OUTCOME

The V1b notes specify 200 mm singlets and a nominal lens-plane distance of 202.375 mm from the sample datum. The geometry provides a starting point for physical alignment.

TOOLS & METHODS

SolidWorks · Optomechanics · Fiber coupling · 1115 nm · Layout planning

1115 nm
Design wavelength

200 mm
Selected lens focal length

Commercial optics and cryostat models are manufacturer geometry. The contribution is assembly layout and integration. The build notes identify alignment approximations. Fiber coupling, window properties, and optical performance still require experimental validation.

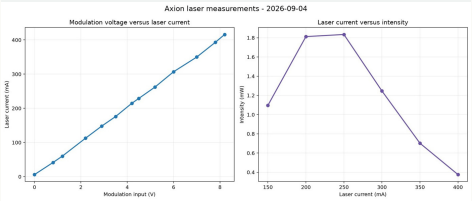
OPTICS / PROJECT 16

Laser collimation and characterization

CAD + BENCH MEASUREMENTS

Laser-assembly and measurement work for the crystal-spectroscopy platform. CAD files capture a beam-collimation assembly, while lab notes and a bench photograph document laser setup and current-modulation characterization.





September 4 bench photograph from the lab notebook. Additional view: Saved modulation/current and intensity plots, labeled September 4, 2026.

CONTRIBUTION

- Modeled a collimation assembly around commercial lens components and a gain-chip reference.
- Set up the laser, drivers, fibers, and wavemeter; recorded current-modulation and optical-intensity measurements.

DELIVERABLES

Beam-collimator assembly revisions · September laser-setup and measurement notes · Bench photograph and saved measurement plot

OUTCOME

The September 4 notebook reports a driver response of 50.1 +/- 4.7 mA/V. The saved intensity sweep also records a turnover near 230 mA for that particular measurement.

TOOLS & METHODS

SolidWorks · Laser alignment · Current drivers · Optical measurement · Lab notebooks

50.1 ± 4.7

mA/V, notebook result

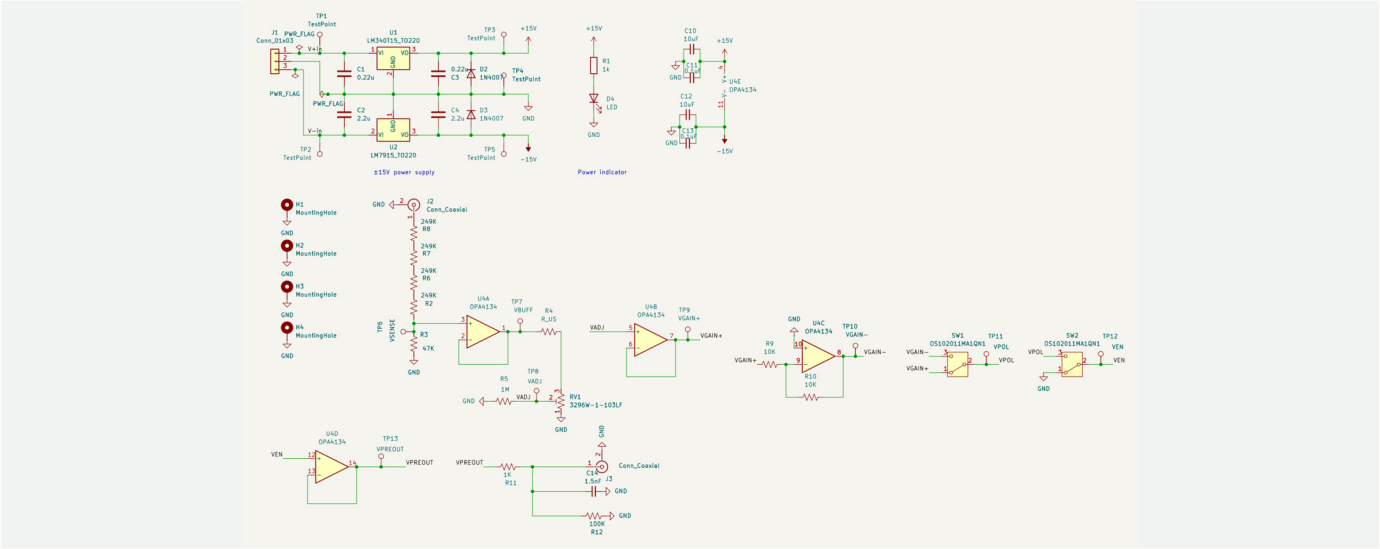
This is setup, integration, and characterization work within the shared Axion crystal project. The driver-response value is a recorded notebook result. Completed linewidth, frequency-stability, or mode-hop-free tuning measurements are not established here.

ELECTRONICS / PROJECT 17

Laser feedforward circuit

CIRCUIT DESIGN / IN PROGRESS

A feedforward circuit project connecting piezo-voltage changes to the laser-current modulation input. The saved schematic and notebook calculations address adjustable gain, polarity, buffering, and calibration of the required transfer function.



Schematic exported from the saved KiCad design; the PCB work file has no valid board outline.

CONTRIBUTION

- Developed a KiCad schematic with an OPA4134-based signal chain, supply regulation, and named test points.
- Derived the gain relationship from piezo tuning, driver response, the divider ratio, and potentiometer setting.

DELIVERABLES

KiCad schematic, PCB work file, and custom footprint · Notebook equations and resistor-value/calibration notes

OUTCOME

The archive captures a useful circuit-development stage: a detailed schematic and transfer-function calculation with the final calibration-dependent resistor value still open.

TOOLS & METHODS

KiCad · Analog electronics · Feedforward · Calibration · Transfer functions

OPA4134

Signal-chain component

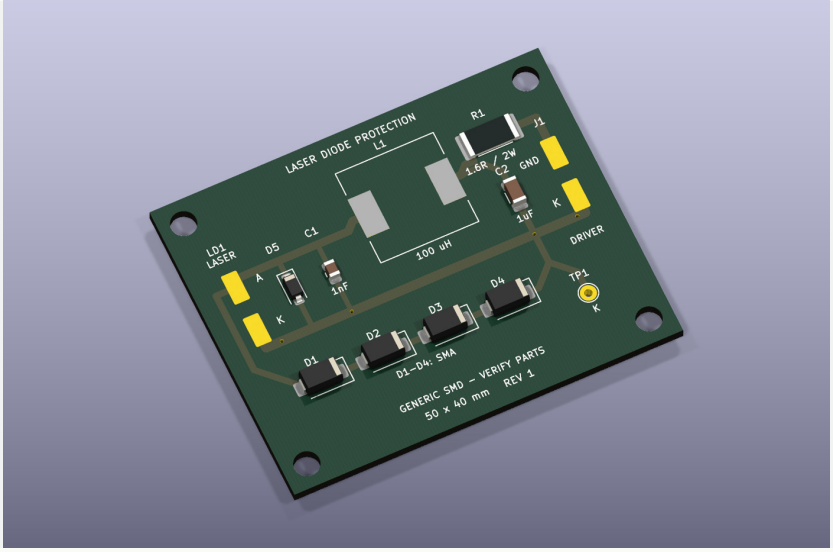
The project includes inherited lab circuit techniques and external laser-system references; this presentation focuses on the saved adaptation and calculations. The saved PCB is unfinished: native rendering reports a missing or malformed outline. No fabricated or validated feedforward board is claimed.

ELECTRONICS / PROJECT 18

Laser-diode protection PCB

ADAPTED SCHEMATIC + PCB

A compact surface-mount implementation of a published laser-diode protection circuit. The KiCad project preserves the source reference and makes the external laser, driver, and cathode test-point connections explicit.



3D rendering of the saved 50 x 40 mm PCB. Some component geometry is schematic or absent.

CONTRIBUTION

- Recreated the thesis schematic and laid out a 50 by 40 mm PCB with labeled external pads.
- Documented the off-board laser connection, driver connection, and component/package checks in the design.

DELIVERABLES

KiCad schematic and routed PCB · Custom solder-pad footprint and symbol library

OUTCOME

The design artifact translates a published protection network into a connector-free board layout that can be reviewed alongside its real-world wiring.

TOOLS & METHODS

KiCad · SMD layout · Circuit adaptation · External wiring

50 × 40 mm

Saved board size

Circuit source: Eric B. Norrgard thesis, Appendix G, Figure G.2, printed page 270; explicitly credited in the saved schematic. The design specifies generic SMD packages and calls for part-rating, pinout, and housing-bond verification. Fabrication and protection performance are not established.

RESEARCH / PROJECT 19

Crystal-spectroscopy platform planning

RESEARCH / SYSTEM PLANNING

Systems research supporting the Tm2+:CaF2 crystal project: translating the scientific goal into requirements for the first laser, cryogenic environment, optical access, and material-characterization instruments.



Portfolio roadmap synthesized from the saved planning documents. Future experimental stages are labeled as planned.

CONTRIBUTION

- Organized literature, project requirements, equipment comparisons, and a staged laser-characterization plan.
- Compared cryostat constraints and UV-Vis-NIR spectroscopy options while separating prototype goals from the long-term detector concept.

DELIVERABLES

Compiled laser, cryostat, and spectrometer planning documents · Literature and manufacturer-reference library · Original project proposals retained as background

OUTCOME

The planning record identifies a first spectroscopy stage near 1116 nm and the measurements needed to characterize it, providing context for the optical CAD and bench work elsewhere in this portfolio.

TOOLS & METHODS

Literature review · Systems engineering · Optical spectroscopy · Requirements · Documentation

~1116 nm
First-stage laser target

3 subsystems
Laser, cryostat, spectroscopy

The detector concept, original proposals, journal papers, and manufacturer documents are background sources. Compiled chat/planning notes are treated as planning evidence, not experimental validation. No completed dark-sector detector, instrument purchase, crystal-spectrum result, or validated cryogenic performance is claimed by this planning case study.

Across the work

I connect scientific requirements with practical hardware, readable software, and clear records of what was designed, built, or measured.

Mechanical design and fabrication

SolidWorks assemblies, fixtures, instrument packaging, waterjet drawings, 3D-print preparation, fit revisions, and parts lists.

Scientific software and analysis

Python interfaces, modular device integration, acquisition threads, HDF5/CSV storage, numerical ODE models, weighted fitting, and saved analysis notebooks.

Electronics and optical integration

KiCad design, connector and enclosure interfaces, circuit adaptation, laser setup, optical layout, and bench characterization.

ABOUT THE RECORD

Prepared from the supplied Lab work archive, reviewed October 27, 2026. Related versions are grouped into project families. Commercial parts, inherited laboratory software, and published circuits are credited where relevant. CAD designs, simulations, research plans, and recorded measurements retain their distinct status.

The companion Source_Index.md gives the full source register. Portfolio_Content.md provides editable text. The offline edition contains the same project stories with searchable galleries.

Brendan P. Hellman

bhellma22@gmail.com